

# Tzu-Ming Harry Hsu

徐子旻

## Research-to-product ML builder for embodied AI

*Embodied AI, ML systems, and AI product delivery*

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- MIT-trained ML researcher-engineer with 10+ years across computer vision, federated learning, sensing, and production AI systems.
- Built and scaled ML/data platforms from research prototypes to deployed systems, including 10 TB data operations, labeling pipelines, and large-scale training infrastructure.
- Recent work spans AI-assisted dental design and sensor-driven human performance products; now targeting general-purpose robotics and embodied AI.
- Founder-operator experience raising \$1MM+, hiring and leading teams up to 20 people, and translating technical roadmaps into product and commercialization plans.

## Education

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### Massachusetts Institute of Technology

*Ph.D., Computer Science*

Jun 2020 – Jun 2022

Cambridge, MA

- *Advisor:* Peter Szolovits
- *Research areas:* Computer vision, federated learning, machine learning for healthcare

### Massachusetts Institute of Technology

*S.M., Electrical Engineering and Computer Science*

Sep 2017 – May 2020

Cambridge, MA

- *Advisors:* Peter Szolovits, Fadel Adib
- *Research areas:* Computer vision, machine learning, wireless signal, signal processing

### National Taiwan University

*B.S., Physics; B.S.E., Electrical Engineering*

Sep 2011 – Jun 2016

Taipei, Taiwan

- *Class rank:* 1/190
- *GPA:* 3.99 / 4.00

## Professional Experience

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### Dentscape

*Machine Learning & Software Engineering Consultant*

Apr 2025 – Present

San Francisco, CA / Taipei, Taiwan

Led ML platform, data, and deployment upgrades for AI-assisted dental design, cutting data access from ~3 days to minutes, reducing labeling turnaround from ~4 weeks to ~1 week, improving backend throughput by ~110%, and lowering infrastructure cost by ~50%.

- Reorganized a fragmented ~10 TB dental dataset into a standardized ~4 TB ML-ready registry, reducing raw-data access from ~3 days to minutes.
- Rebuilt labeling and pre-labeling workflows end-to-end, cutting 100-200 case batch turnaround from ~4 weeks to ~1 week with clear audit trails.
- Standardized devops/deployment with modern packaging, private registry support, and GitHub Actions; deploys moved from manual half-day ops to single-push workflows.
- Led infra migration from GPU-heavy services to CPU-compatible stack for CPU-bound workloads, reducing overall server costs by ~50%.
- Implemented bilevel/meta-learning framework for personal AI crown generation and improved preference fit by ~20%, contributing to higher user engagement.
- Optimized backend pipelines for SLO segmentation and crown generation, achieving ~110% aggregate throughput improvement.

- Introduced structured hiring and engineering standards; closed a senior leadership hire and supported award-related media visibility.

## **Clarq AI**

*Co-founder & CTO*

Dec 2024 – Present

*Palo Alto, CA / Taipei, Taiwan*

Helped reposition Clarq from a hardware product into a force-intelligence platform with 100+ deployments, 15M+ force-time samples, and 150K labeled motion sets, linking sensor data strategy to product and fundraising.

- Co-led technical positioning around a three-layer strategy: hardware wedge, SaaS profit engine, and long-term data/model moat.
- Drove US-facing investor outreach and progressed pre-seed conversations toward near-commit stage with angels.
- Supported Startup Island Taiwan Silicon Valley program participation and international expansion narrative.
- Aligned product roadmap, data strategy, and investor-facing story across Taiwan hardware operations and US capital market expectations.

## **Massachusetts Institute of Technology**

*Research Assistant*

Sep 2018 – Jun 2022

*Cambridge, MA*

Led research on deployable medical AI and computer vision, producing 20+ publications, patents, and theses while collaborating with Google, Takeda, and hospital partners to move models toward real workflows.

- Published 14 conference articles, 5 journal articles, one U.S. patent, and two degree theses.
- Served as a peer reviewer for 8 conferences and journals, reviewing 34 articles.
- Received media coverage from CBS, Engadget, TechCrunch, and MIT News.
- Collaborated with Google, Takeda Pharmaceutical, and Beth Israel Deaconess Medical Center.
- Awarded scholarships totaling 4 years in duration.

## **Google**

*Student Researcher*

Jun 2019 – Mar 2020

*Seattle, WA / Cambridge, MA*

Built large-scale ML experimentation infrastructure spanning 1,000+ GPUs and 1M+ machine hours, and published 2 peer-reviewed papers in deep federated learning with open-source outputs for broader adoption.

- Conducted foundational research in deep federated learning, contributing to early growth of the field.
- Published 2 peer-reviewed papers and presented findings at top conferences and workshops.
- Developed a scalable experimentation library running on thousands of machines and 1,000+ GPUs (over 1M machine hours).
- Open-sourced research code and datasets for broader community use.

## **CodeGreen Labs PBC**

*Co-founder & CTO*

Sep 2023 – Nov 2024

*Boston, MA / Taipei, Taiwan*

Led a 10-person engineering organization delivering auditable data systems for enterprise partners, reaching projected \$1MM annual revenue while hardening integration, compliance, and delivery operations.

- Reached projected annual revenue of \$1MM USD by Q2 2024 with a 10-person team.
- Established partnerships with Microsoft, Global Blockchain Business Council, Gold Standard, SDG Data Alliance, and PVBLC.
- Built a cross-functional team across product, partnerships, and engineering to address complex data, auditability, and compliance challenges.
- Designed and shipped data-intensive systems with end-to-end auditability, regulatory compliance, and interoperability with existing partner ecosystems.

## **Hashgreen Labs**

*Co-founder & CTO*

Sep 2022 – Aug 2023

*Boston, MA / Taipei, Taiwan*

Built and led a 20-person engineering organization, raised \$1MM+ in capital, and shipped client-facing data and transaction systems while establishing hiring, delivery, and product operating cadence from zero.

- Raised over \$1MM USD in capital and grew the team from 0 to 20 employees.

- Built hiring, onboarding, and performance review systems to support a high-performing engineering organization.
- Secured and delivered high-impact projects for clients including The World Bank and Chia Network.
- Led development of DeFi protocols (DEX and AMM), becoming the first team on the Chia blockchain to launch these primitives.

## **International Academia of Biomedical Innovation Technology**

2021 – Present

*Consultant & Convener*

*Taipei, Taiwan*

Supported a 50+ member biomedical innovation community through project mentorship, dental AI research collaboration, and large-scale technology outreach events.

- Helped grow a community of 50+ members while supporting 8+ academic projects and 15+ publications.
- Led a dental AI research team in collaboration with National Taiwan University Hospital.
- Co-organized demo events with 100+ participants to showcase biomedical technologies and educate the broader community.

## **WorldQuant**

Jun 2021 – Aug 2021

*Data Science Intern*

*Taipei, Taiwan*

Improved an existing quantitative strategy by +1.5% market correlation through targeted modeling and optimization work.

## **Digital Drift**

Mar 2016 – Mar 2020

*Lead Machine Learning Scientist*

*Taipei, Taiwan*

Built and deployed deep learning systems and backend APIs that moved computer vision features from experimentation into production mobile experiences.

- Built deep neural networks for cuisine recognition and matching using TensorFlow on multi-GPU machines.
- Designed and maintained backend APIs to serve model predictions in production settings.

## **Google**

Jun 2020 – Sep 2020

*Software Engineer Intern*

*Remote*

Contributed production software and experimentation tooling that improved reliability and iteration speed for applied machine learning workflows.

## **Brigham and Women's Hospital**

Mar 2019 – Feb 2020

*Research Trainee*

*Boston, MA*

Contributed hospital-facing medical-imaging ML work on risk prediction and deployment-oriented evaluation using real patient cohorts and radiology workflows.

- Supported model development and validation analysis for clinical imaging applications tied to operational hospital use cases.

## **Beth Israel Deaconess Medical Center**

Nov 2019 – Oct 2020

*Research Trainee*

*Boston, MA*

Published impactful clinical AI research and deployed a PACS-integrated screening workflow across 1,000+ patients while managing shared deep-learning compute for a 10+ member research unit.

- Published three impactful research articles on critical medical topics, including liver tumor assessment, COVID-19 clinical risk prediction, and automatic pancreatic cancer assessment.
- Developed and implemented an AI-driven patient screening pipeline in the hospital's PACS workflow, screening over 1,000 patients and improving disease diagnosis accuracy.
- Managed the computation resources for a research unit of 10+ members, streamlining deep learning processes and maximizing resource sharing.

## **Academia Sinica**

Feb 2014 – May 2016

*Student Researcher*

*Taipei City, Taiwan*

Researched heterogeneous, unsupervised, and semi-supervised domain adaptation for visual classification and published three peer-reviewed papers presented at top conferences and workshops.

- Researched heterogeneous, unsupervised, and semi-supervised domain adaptation for visual classification.
- Published 3 peer-reviewed academic papers and presented findings at top conferences and workshops.

### **Olympiad Tutoring Community**

Sep 2011 – Jun 2015

*Private Tutor*

*Taipei, Taiwan*

Mentored advanced physics students from fundamentals through competition-level performance, including pathways to Taiwan's national Olympiad representation.

- Mentored students who became Taiwan national representatives in the International Physics Olympiad (IPhO).

### *Leadership Experiences*

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#### **Federation of Taiwanese Student Associations in New England (FT-SANE)**

2019 – 2020

*Activities Officer*

*Boston, MA*

#### **MIT Taiwanese Student Association**

2018 – 2019

*President*

*Boston, MA*

- Led coordination of 20+ events for a diverse community of over 100 members.
- Organized 10+ career workshops and helped plan 10+ national holiday celebrations.

### *Awards & Honors*

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#### **Departmental Valedictorian (First Place at Graduation)**

2016

*Department of Electrical Engineering, National Taiwan University*

*Taipei, Taiwan*

- Placed 1st among 200 classmates across all courses.

#### **Silver Medal**

2015

*Altera Innovate Asia FPGA Design Competition*

*Taipei, Taiwan*

- Placed 2nd among 20 international teams.
- Designed a product named EZBud with integrated algorithms that modulate music based on user sporting statistics.

#### **Gold Medal / Overall Winner**

2011

*42nd International Physics Olympiad (IPhO)*

*Bangkok, Thailand*

- Represented Taiwan and placed 1st in theory, experiment, and combined rankings among 400+ participants from 80+ countries.

#### **Gold Medal / Experiment Winner**

2011

*12th Asian Physics Olympiad (APhO)*

*Tel Aviv, Israel*

- Placed 1st in the experiment section among 100+ participants.

#### **Gold Medal**

2010

*41st International Physics Olympiad (IPhO)*

*Zagreb, Croatia*

#### **Gold Medal / Experiment Winner**

2010

*11th Asian Physics Olympiad (APhO)*

*Taipei, Taiwan*

- Placed 1st in the experiment section among 100+ participants.

#### **Gold Medal**

2008

*5th International Junior Science Olympiad (IJSO)*

*Changwon, Korea*

**Artificial Intelligence to Assess Dental Findings from Panoramic Radiographs—A Multinational Study** 2025

*arXiv*

*Cited by 8*

Yin-Chih Chelsea Wang, Tsao-Lun Chen, Shankeeth Vinayahalingam, Tai-Hsien Wu, Chu Wei Chang, Hsuan Hao Chang, Hung-Jen Wei, Mu-Hsiung Chen, Ching-Chang Ko, David Anssari Moin, Bram van Ginneken, Tong Xi, Hsiao-Cheng Tsai, Min-Huey Chen, **Tzu-Ming Harry Hsu**, Hye Chou

**Intra-Oral Scan Segmentation Using Deep Learning** 2023

*BMC Oral Health*

*Cited by 42*

Shankeeth Vinayahalingam, Steven Kempers, Julian Schoep, **Tzu-Ming Harry Hsu**, David Anssari Moin, Bram van Ginneken, Tabea Flugge, Marcel Hanisch, Tong Xi

**DeepOPG: Improving Orthopantomogram Finding Summarization with Weak Supervision** 2021

*Medical Image Computing and Computer Assisted Intervention (MICCAI 2021)*

*Cited by 8*

**Tzu-Ming Harry Hsu**, Yin-Chih Chelsea Wang

**Federated Visual Classification with Real-World Data Distribution** 2020

*European Conference on Computer Vision (ECCV 2020)*

*Cited by 275*

**Tzu-Ming Harry Hsu**, Hang Qi, Matthew Brown

**Learning Food Quality and Safety from Wireless Stickers** 2018

*Proceedings of the 17th ACM Workshop on Hot Topics in Networks (HotNets 2018)*

*Cited by 95*

Unsoo Ha, Yunfei Ma, Zexuan Zhong, **Tzu-Ming Hsu**, Fadel Adib

**3D-Aware Scene Manipulation via Inverse Graphics** 2018

*Advances in Neural Information Processing Systems (NeurIPS 2018)*

*Cited by 321*

Shunyu Yao, **Tzu Ming Hsu**, Jun-Yan Zhu, Jiajun Wu, Antonio Torralba, Bill Freeman, Josh Tenenbaum